KLH Bachupally

AI-Based Phishing Detection System Sequence Learning and Safe Context Analysis

CharCNN + BiLSTM + Attention
Production v3.3 and Domain-Separated Research Candidate v3.4

Students — B.Tech Project • KLH CSIT

Innovation — Robustness • Context • Verified Learning

Research — Versioned Data and Evaluation

Impact — Real-Time Detection • FastAPI • Web Demo

Python | PyTorch | LightGBM | FastAPI | scikit-learn | Apple MPS | Vercel | 91 tests | GitHub

Abstract

Phishing detection needs reliable evaluation as well as accurate predictions. Our initial 20,000-URL CNN-LightGBM reproduction remains a historical experiment. Production now serves CharCNN with a two-layer bidirectional LSTM and additive attention (v3.3). Its split audit found domain overlap, two cleaned training-input matches in an evaluation snapshot, and a failed promotion criterion overridden by the maintainer. A separate v3.4 candidate repairs whole-domain separation and achieves 92.876% accuracy, 90.696% recall and 98.289% ROC-AUC on 31,795 test URLs. However, fewer external false alarms accompany more missed phishing URLs, so it was not promoted. Independent HTML/email evidence and human-reviewed feedback remain separate from the URL-model score.

Objectives

1. Detect phishing using learned patterns and engineered features
2. Prevent leakage through root-domain-separated evaluation
3. Compare historical ensembles with bidirectional sequence models
4. Measure adversarial resilience and false-positive tradeoffs
5. Add safe context evidence and human-verified adaptation

Historical split: 14,002 / 2,000 / 3,998 — **Historical overlap: 0**

Key Innovations

Version-Specific Evaluation
Original reproduction: zero domain overlap. v3.3 has overlap; the local v3.4 candidate restores whole-domain separation.

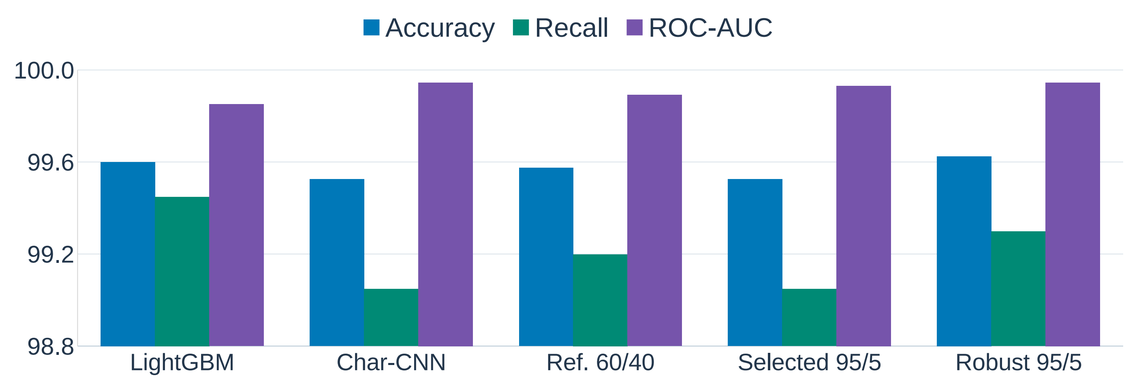
Adversarial Robustness
Historical robustness experiment: eight seeded mutations, 1,751 train-only augmentations; recall 98.145% to 99.148%.

Safe Context Analysis
Independent HTML and email analyzers inspect supplied text. No website fetching, code execution or mailbox access.

Human-Verified Continuous Learning
Quarantine → human review → approved data → candidate training → validation gate. Model registry and PSI drift reports.

Historical Experimental Results

Model Performance Comparison



Percent; expanded 98.8–100 axis. Historical clean test: 3,998 URLs.

100%	0	37 → 17	91 passed
Original precision	Historical overlap	Adversarial misses	Production tests

Proposed Methodology / System Architecture

URL String and Shared Canonicalization
Trim spaces; normalize scheme/host; equate root slash only

↓

Character Encoding and Embedding
First 256 characters; 97 indices; embedding dimension 64

↓

Parallel CharCNN: Kernels 3 / 5 / 7, 128 Channels Each
BatchNorm + ReLU; MaxPool(2) gives 128 time steps

↓

Two-Layer Bidirectional LSTM
Input 384; hidden 128 per direction; recurrent dropout 0.2

↓

Additive Attention + Temporal Max Pool
512 values; dense head 128 -> 32 -> 1

↓

Sigmoid and Fixed Threshold 0.5
v3.3 deployed; v3.4 research candidate NOT promoted

Legacy CNN-LightGBM ensemble scores remain reference-only.

Research Extensions

A. Historical Adversarial Robustness
Train URLs → 8 mutations → Robust fit → Val. gate → Candidate

B. Independent Context Analyzers
HTML text — Forms, passwords, scripts, iframes

Email text — Risk terms, requests, URLs

Supplementary evidence only; excluded from validated URL probability

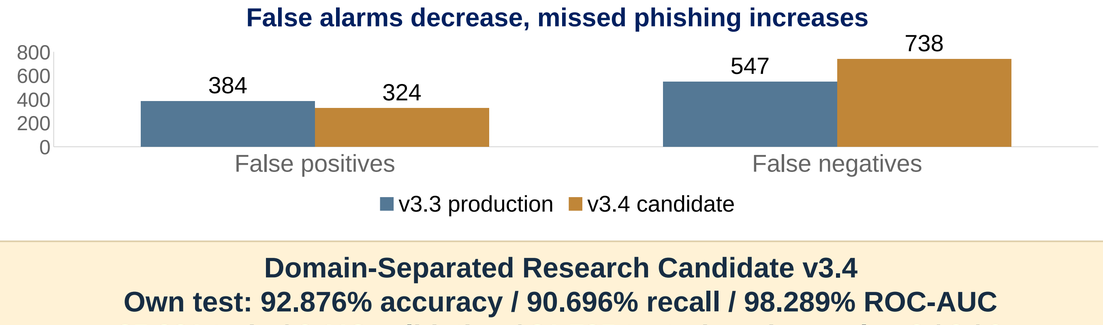
C. Human-Verified Adaptation
Feedback → Quarantine → Human review → Approved data
Offline candidate training → Validation gate → Model registry

PSI drift monitoring: monitor ≥ 0.10; investigate ≥ 0.25. No automatic retraining.

Current Production vs Research Candidate

External snapshot: same 11,271 inputs; different training histories

False alarms decrease, missed phishing increases



Domain-Separated Research Candidate v3.4
Own test: 92.876% accuracy / 90.696% recall / 98.289% ROC-AUC
25,961 train / 6,113 validation / 31,795 test; domain overlap 0 / 0 / 0

Promotion gate: FAILED. Production remains v3.3.
Previously inspected snapshots do not establish blind temporal generalization.

Historical Reference and Reproduction

Study / model	URLs	Accuracy	Precision	Recall	ROC-AUC
Reference study [1]	99,361	99.819%	100%	99.635%	99.947%
Our reproduction	20,000	99.525%	100%	99.048%	99.931%
Our robust candidate	**20,000***	**99.625%**	**99.950%**	**99.298%**	**99.945%**

*20,000 base URLs + 1,751 train-only mutations. Different study protocols.

Current v3.3: 91.740% external accuracy; 91.277% recall.
External snapshot: 11,271 API-compatible rows, previously inspected.
91 production-source tests passed; research-branch tests are separate.

References

[1] R. Dubey, A. M. Tripathi, A. Srivastava & S. Singh. “Phishing Detection System: An Ensemble Approach Using Character-Level CNN and Feature Engineering.” arXiv:2512.16717, 2025.
[2] PhishTank. Valid Phishing URLs. phishtank.org.
[3] Tranco. Top 1M List. tranco-list.eu.
[4] H. Le et al. “URLNet.” arXiv:1802.03162, 2018.
[5] P. Yang, G. Zhao & P. Zeng. “Phishing Website Detection Based on Multidimensional Features Driven by Deep Learning.” IEEE Access 7, 2019.
[6] S. Garera et al. “A Framework for Detection and Measurement of Phishing Attacks.” ACM WORM, 2007.

Team Details

Guide
Dr. K Venkateshwara Rao

Team Members
2320090017 — D Sunidhi
2320090060 — P Manvitha
2320090069 — G Likith

KLH Bachupally